Civics Group	Index Number	Name (use BLOCK LETTERS)

H1

ST. ANDREW'S JUNIOR COLLEGE
2014 Preliminary Examination

H1 BIOLOGY **8875/2**

Paper 2: Core (Mark Scheme)

Tuesday 2 September 2014 2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Compulsory question to be answered on writing paper provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/9½
2	/8
3	/10
4	/12½
5	/20
Total	/60

This document consists of **18** printed pages.

Section A

Answer **all** *questions*.

QUESTION 1

The fruitfly, *Drosophila*, has many different species. Three of these species, *Drosophila pseudoobscura*, *D. persimilis* and *D. miranda*, are thought to be closely related.

Samples of these three species were collected from the western United States of America.

Fig. 1.1 shows where these species naturally occur.

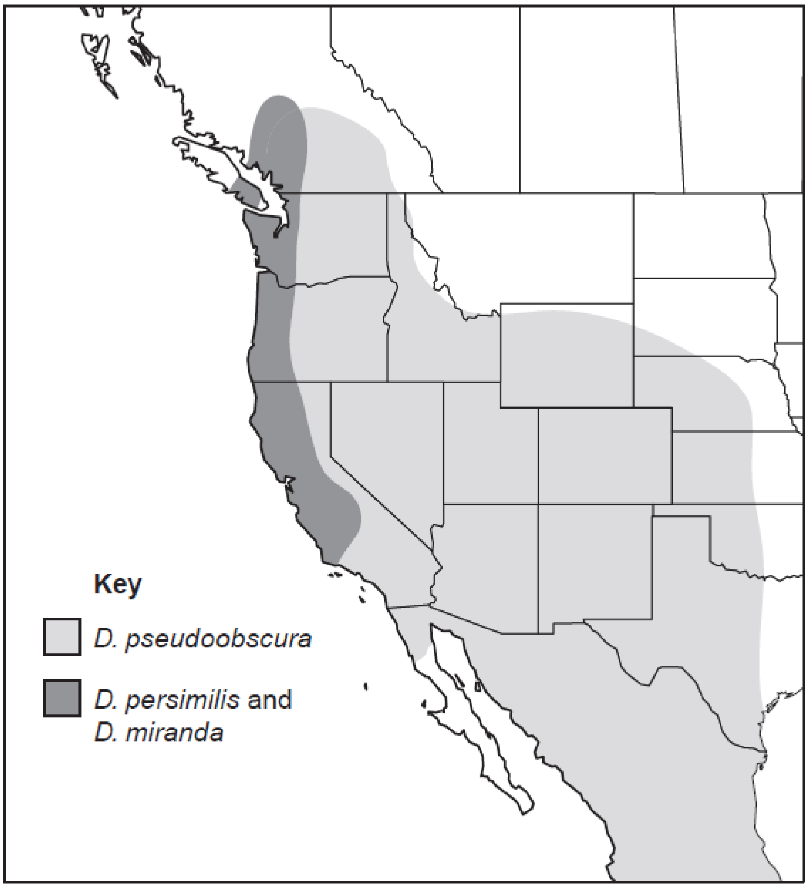

Fig. 1.1

The base sequences of four regions of DNA of each species were sequenced. The divergence of these base sequences in *D. pseudoobscura* and *D. persimilis* from the sequences in *D. miranda* was calculated. The results are shown in Table 1.2.

DNA region	*Drosophila* species	percentage divergence of base sequence from that of *D. miranda*
1	*pseudoobscura*	2.5
	persimilis	2.4
2	*pseudoobscura*	8.1
	persimilis	7.3
3	*pseudoobscura*	2.1
	persimilis	1.7
4	*pseudoobscura*	1.9
	persimilis	1.7

Table 1.2

(a) With reference to Table 1.2, describe the evidence that *D.miranda* may be more closely related to *D.persimilis* than to *D.pseudoobscura.*

…………………………………………………………………..……………………...[1½]

1 Percentage divergence values of *D.persimilis* from *D.miranda* is less than that of *D.pseudoobscura* from *D.miranda*
/ ora ;

2 For <u>all/4</u> DNA regions ;

3 Use of appropriate figures ;
/ e.g. at DNA region 1, percentage divergence of *D.persimilis* is 1.7 and 1.9 for pseudoobscura

(b) Suggest why there is more divergence in some regions of DNA than in others.

…………………………………………………………………..………………………...[1]

1 Some regions of DNA have <u>higher mutation rates</u> / <u>more prone to mutations</u> ;

2 (Mutational changes) are Less harmful when exact sequence of amino acid is not critical to the survival of the organism ;

3 Lower divergence / mutation rates if DNA is part of an <u>important</u> gene ;

4 Mutations in some regions are likely to be fatal (so not seen in populations) ;

(c) The area where D. pseudoobscura is found is separated from the areas where the other two species are found by a high range of mountains.

Explain how the species D. pseudoobscura could have evolved from a population of D. miranda.

……………………………………………………………………..………………………..[4]

1 Geographical isolation due to physical barrier / high mountains ;
2 Disruption to gene flow / no interbreeding between two ancestral populations of *D.miranda* ;

3 Phenotypic variations exist in the two populations of *D.miranda* ;
REJECT ***D.miranda* isolated from *D.psuedoobscura***
4 Genetic variations exist due to mutations ;

5 Natural selection occur ; CREDIT **only if presented in correct context**
6 Different selection pressures in different environmental conditions ;
7 Those with advantageous alleles survive to reproductive age and pass down alleles to offspring ;
8 Genetic drift occur ; CREDIT **only if in presented in correct context**
9 (5 and 7) Lead to changes in frequency of alleles in the gene pool ;

10 Accumulation of genetic differences over time ;
11 Two populations ultimately cannot interbreed to produce viable, fertile offspring ;
12 Two populations reproductively isolated ;
13 New species (*D.psuedoobscura*) formed by allopatric speciation.

(d) Besides the use of DNA sequences to analyse evolutionary relationships between species, anatomical homologies are also used.

Explain how anatomical homologies observed in the forelimbs of the animals shown in Fig. 1.3 support Darwin's theory of natural selection.

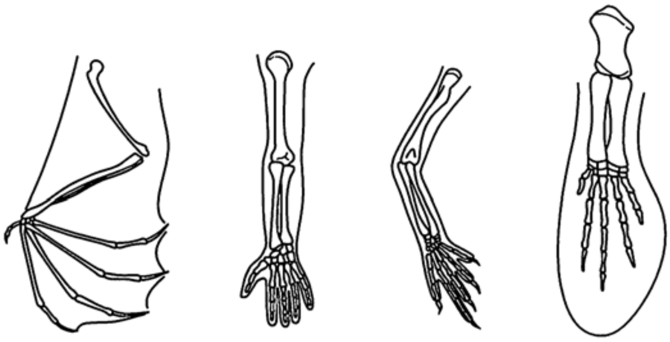

Fig.1.3

……………………………………………………………………..……………………….. [3]

"Descent"

1 Same <u>basic arrangement of bones</u>
/ each forelimb has a humerus, radius, ulna, carpals, set of phalanges

2 Inherited from common ancestor ;

"With modifications"

3 Forelimbs show considerable differences in <u>form and function</u>
/ Form of mammalian forelimbs modified to perform a variety of functions e.g. swimming, flight, crawling, etc ;

4 Due to differences in selection pressures faced by different organisms ;

5 Causes changes in genes and allele frequencies in gene pool ;

6 Shows evolutionary divergence
/ divergent evolution
/ branching out of many species, from a common ancestor in adaptive radiation ;

[Total : 9½]

QUESTION 2

A student carried out an investigation involving uptake of the stain methylene blue by yeast cells.

The investigation involved adding methylene blue to a suspension of yeast cells. Samples of the stained yeast cells were heated to different temperatures.

The student then observed the cells at high power under a light microscope.

The results are shown in Table 2.1.

temperature (°C)	cells observed stained blue (%)	colour of solution surrounding cells
10	98	colourless
20	96	colourless
30	97	colourless
40	96	colourless
50	73	colourless
60	12	light blue
70	2	blue
80	0	blue

Table 2.1

(a) Yeast cells take up methylene blue by active transport. Using only the information provided in Table 2.1, outline and explain the evidence that supports this statement.

……………………………………………………………………………..……………………....[3½]

1 [Trend observed] At low temperatures, all stain is in the cells
[Supporting data] Between 10^{0}C to 50^{0}C, percentage of stained cells is between 73% to 98% ;
2 [Explanation] Taken up against concentration gradient ;

3 [Trend observed] At high temperatures, stain is not found in the cells ;
4 [Supporting data] From 60 to 80^{0}C, percentage of stained cells decreases from 12% to 0% ;
5 [Supporting data] Surroundings of cells stained blue ;
6 [Explanation] Enzymes denatured so <u>no ATP</u> for active transport (of stain) ;
REJECT **'enzyme denatured' alone**
CREDIT **pumps /carrier, proteins involved in active transport denatured**
ACCEPT **mitochondria non-functional, no ATP produced**

7 [Supporting data] Use of comparative figure e.g. 98% at 10^{0}C and 0% at 80^{0}C

(b) Suggest why some cells did not stained blue at 20^0C.

……………………………………………………………………………..………………………......[1]

1 **Cells dead / not respiring ; REJECT 'burst'**
2 **<u>No ATP</u> to hydrolyze for use in active transport ;**
ACCEPT no functioning mitochondria for ATP production ;
REJECT references to energy being produced for active transport ;

(c) Suggest one change that occurred to the cell surface membranes of the yeast cells at temperatures above 60^0C.

……………………………………………………………………………..……………………….....[½]

1 **Phospholipid bilayer, melts / more fluid ;**
2 **Membrane becomes disrupted REJECT 'membrane disintegrate / denature'**
3 **Membrane proteins / carrier molecules, denatured / unable to function ;**

(d) Explain why the stained yeast cells lost their colour at higher temperatures.

……………………………………………………………………………..……………………....[1½]

1 **Membrane <u>permeable</u> to stain ;**
2 **Methylene blue leaked out of cell / released to solution ;**
3 **Movement via diffusion /down concentration gradient.**

The student placed a small sample of the yeast suspension on a microscope slide and observed it under high power.

Fig.2.2 shows what the student observed.

Fig. 2.2

Cell Z is undergoing a process known as budding. Budding is a form of asexual reproduction.

(e) Explain the importance of mitosis in budding.

……………………………………………………………………………..……………………......[1½]

1 **Nucleus formed after mitosis**
/ cells formed after mitosis and subsequent cell division, is genetically identical to the parental cell ;
2 **Ref same number of chromosomes and same DNA sequence ;**
3 **Retains advantage of the organism in adapting to its environment ;**
4 **Ensures continued survival of the species.**

[Total : 8]

QUESTION 3

Fig.3.1 shows the relationship between various metabolic processes.

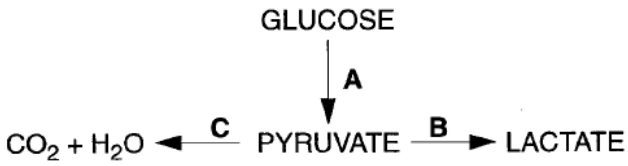

Fig. 3.1

(a) (i) Identify the three metabolic processes.

……………………………………………………………………..………………………...[1½]

1 A – glycolysis ;
2 B – anaerobic respiration / lactate fermentation ;
3 C – Krebs cyclc AND oxidative phosphorylation

(ii) State the letter of the pathway in which acetyl coenzyme A is involved.

…………………………………………………………………..……………………………...[½]

1 C.

(iii) State the letter of the pathway in which ATP is utilized.

…………………………………………………………………..……………………………...[½]

2 A.

In an investigation, mammalian liver cells were homogenized and the resulting homogenate centrifuged. Portions containing only nuclei, ribosomes, mitochondria and cytosol were each isolated. Samples of each portion, and of the complete homogenate, were incubated in four ways,

1 with glucose
2 with pyruvate
3 with glucose plus cyanide
4 with pyruvate plus cyanide

Cyanide inhibits carriers in the electron transport chain, such as cytochromes. After incubation, the presence or absence of carbon dioxide and lactate in each sample were determined.

The results are summarised in Table 3.2.

	samples of homogenate									
	complete		nuclei only		ribosomes only		mitochondria only		cytosol	
	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate
1. glucose	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓
2. pyruvate	✓	✓	✗	✗	✗	✗	✓	✗	✗	✓
3. glucose and cyanide	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓
4. pyruvate and cyanide	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓

✗ = absent ✓ = present

Table. 3.2

(b) (i) With reference to this investigation, name two organelles not involved in respiration.

…………………………………………………………………..………………………....[1]

1 Ribosome ;
2 Nuclei. BOTH for mark

(ii) Explain why carbon dioxide is produced when mitochondria are incubated with pyruvate but not when incubated with glucose.

…………………………………………………………………..……………………....[2½]

1 Pyruvate is end product of glycolysis (in cytosol) ;
2 Pyruvate can enter mitochondria for link reaction and Krebs cycle ;

3 Carbon dioxide produced in link reaction AND Krebs cycle ;
4 Decarboxylation reactions occur in the mitochondria ;

5 Glycolysis does not occur in the mitochondria
/ enzymes involved in glycolysis not present in the mitochondria ;
6 No carriers for glucose in mitochondrial membranes.

(iii) Explain why in the presence of cyanide, lactate production does occur, but not carbon dioxide production.

………………………………………………………………………...……………………....[3]

1 **Cyanide prevents oxidative phosphorylation from occurring ;**
2 **(non-competitive) inhibition of cytochrome oxidase ;**
3 **Reduced NAD / NADH, not oxidised ;**
4 **No decarboxylation occurs in, mitochondria / Krebs cycle and link reaction ;**

5 **Alternative H acceptor needed**
/ pyruvate is H acceptor
/ NADH produced in glycolysis reduce pyruvate ;
6 **To form lactate during anerobic respiration ;**
7 **Does not involved cytochromes ;**
/ Enzymes in cytosol not inhibited by cyanide.

This investigation may be repeated using yeast cells instead of liver cells.

(c) State the products that would be formed by the incubation of glucose with cytosol from yeast.

………………………………………………………………………...……………………....[1]

1 **Carbon dioxide ;**
2 **Ethanol.** BOTH for mark

[Total : 10]

QUESTION 4

(a) In the fruit fly shown in Fig.4.1, *Drosophila melanogaster*, the gene for eye colour is sex-linked. The allele for red eye (R) is dominant to the allele for white eye (r). A student carried out two crosses.

Fig.4.1

The first cross involved crossing a red-eyed female with a white-eyed male. Two fruit flies of the F1 generation were then crossed. The following F2 phenotypes were produced 95 red-eyed females, 49 red-eyed males, 56 white-eyed males.

In the second cross, the student carried out a reciprocal cross.

(i) Draw a genetic diagram in the space below to show the expected F1 and F2 generations from the reciprocal cross.

……[4½]

Reciprocal cross

Parental phenotypes:	Male fly Red-eyed	x	Female fly, White-eyed
Parental genotype:	X^R Y		X^r X^r
Parental gametes :	X^R Y		X^r X^r

Punnett square :

	X^r	X^r
X^R	X^R X^r Female fly, Red-eyed	X^R X^r Female fly, Red-eyed
Y	X^r Y Male fly, White-eyed	X^r Y Male fly, White-eyed

F1 genotype :	X^R X^r		X^r Y
F1 genotype :	Female fly Red-eyed		Male fly White-eyed
F1 genotype :	1	:	1

<u>F1 cross</u>

F1 phenotypes:	Female fly Red-eyed	x	Male fly White-eyed
F1 genotype:	$X^R X^r$		$X^r Y$
F1 gametes :	X^R X^r		X^r Y

Punnett square :

	X^R	X^r
X^r	$X^R X^r$	$X^r X^r$
Y	$X^R Y$	$X^r Y$

F2 genotype :	$X^R X^r$	$X^r X^r$	$X^R Y$	$X^r Y$
F2 genotype :	Female fly Red-eyed	Female fly White-eyed	Male fly Red-eyed	Male fly White-eyed
F2 genotype :	1 :	1 :	1 :	1

1 Parental phenotype ; REJECT incomplete phenotype description
e.g. gender not specified or “white”/ “red” without reference to eye colour
2 Parental genotype ;
3 Parental gametes (circled) ;

4 F1 genotype ;
5 F1 genotype correspond to phenotype ;
6 F1 phenotypic ratio ; REJECT incomplete phenotype description

7 F1 gametes (circled) ;
8 F2 genotype ;
9 F2 genotype correspond to phenotype ;
10 F2 phenotypic ratio. REJECT incomplete phenotype description

(b) The giant panda is an endangered species native to South Central China. Due to its conservation status, a lot of research studies have revolved around the giant panda. Recent research on a protein from the giant panda was carried out in 2008. This protein, called crystalline, is found in the lens of the eye, and has a sequence that has been highly conserved in mammals.

The steps in the procedure used in the study are summarized in Fig.4.2.

Step 1	Reverse transcriptase is used to make crystallin cDNA from the panda's mRNA.
Step 2	The panda crystallin cDNA undergoes a process to increase the quantity of cDNA.
Step 3	This cDNA is introduced into bacterial cells using a plasmid and the bacterial cells are grown in Petri dishes.
Step 4	Panda crystallin protein is harvested from the bacteria and is separated from other sized proteins on a gel.
Step 5	The crystallin protein and cDNA are both sequenced by automated sequencing methods.

Fig.4.2

(i) Identify the technique used in each case to carry out steps 2 and 4.

……………………………………………………………………………………………………[1]

1 Step 2 – polymerase chain reaction ;
2 Step 4 – electrophoresis ;

(ii) Explain why lens cells were used for the isolation of the giant panda's mRNA.

……………………………………………………………………………………………………[1]

1 (Only lens cells in panda) expresses the crystalline gene / transcribes the crystalline gene and translated the mRNA coding for the crystalline protein ; REJECT crystalline found in lens
2 Cells are <u>highly enriched in mRNA</u> for crystalline gene ;

(iii) The giant panda crystalline protein cloned was 175 amino acids long, corresponding to a 528 base pair cDNA gene. Explain why a protein that is 175 amino acids long is coded for by 528 base pairs of DNA.

……………………………………………………………………………………………………[1½]

1 Each amino acid is encoded for by a triplet code / 3 bases ;
2 175 amino acids x 3 = 525 base pairs ;
3 3 bases are the stop codons which do not encode for any amino acid

(iv) In step 3 of the procedure shown in Fig.4.2, a plasmid was used in the cloning of the crystalline gene.

On the diagram of the plasmid shown in Fig.4.3, draw two named genetic markers and the other components that are required for the expression vector to be suitable for use in the cloning of the crystalline gene. You are to label these components with specific names where applicable.

……………………………………………………………………..……………….........[3]

1 *__LacZ__* __gene__ ;
2 __Ampicillin-resistance gene__ (or any other named antibiotic resistance gene) ;

OR
1 2 antibiotic resistance __genes__; eg. amp^R or tet^R
2 2 antibiotic resistance __genes__; eg. amp^R or tet^R

3 Origin of replication ;
4 Polylinker/multiple cloning site __within one marker gene__;
5 Named restriction site e.g., *Bam*HI / *Xma*I / *Hin*dIII / *Eco*RI;
6 __bacteria__ promoter before crystalline gene (so that inserted gene can be expressed)

(v) Explain how the plasmid in Fig.4.3 was introduced into bacterial cells in step 3 of the procedure.

……………………………………………………………………..……………….........[1½]

1 Bacteria treated with ice-cold $CaCl_2$ solution and placed on ice ;
2 To make cells competent ;

3 Subjected to brief heat shock period ;
4 To create transient pores created in the cell membranes for recombinant plasmid to enter the bacteria.

[Total : 12½]

Section B

Answer the question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a), (b)** etc., as indicated in the question.

5 **(a)** With reference to sickle cell anemia, explain how a change in the sequence of the DNA nucleotide affects the phenotype of the organism. [7]

(b) Describe the development of cancer as a multistep process. [8]

(c) The time course of an amylase digestion can be followed by measuring the rates of formation of the product, glucose.

Describe how the concentration of glucose produced can be measured using a colorimeter. [5]

(a) With reference to sickle cell anemia, explain how a change in the sequence of the DNA nucleotide affects the phenotype of the organism.
……………………………………………..……………………..……………………..………[7]

1 base substitution in the haemoglobin β chain gene (on chromosome 11) ;
2 e.g. A to T substitution on DNA ;
INDICATE "several allelic variants present, one of the most common form is …"
3 causes a change in mRNA codon ;
4 resulting in change in the sixth amino acid from glutamic acid to valine ;

Table 1: Single-Base Mutation Associated with Sickle-Cell Anemia

Sequence for Wild-Type Hemoglobin												
ATG	GTG	CAC	CTG	ACT	CCT	GAG	GAG	AAG	TCT	GCC	GTT	ACT
Start	Val	His	Leu	Thr	Pro	Glu	Glu	Lys	Ser	Ala	Val	Thr
Sequence for Mutant (Sickle-Cell) Hemoglobin												
ATG	GTG	CAC	CTG	ACT	CCT	GTG	GAG	AAG	TCT	GCC	GTT	ACT
Start	Val	His	Leu	Thr	Pro	Val	Glu	Lys	Ser	Ala	Val	Thr

Credits – Nature publications

5 change in amino acid sequence / primary structure, of HbS protein ;
6 glutamic acid is hydrophilic whereas valine is hydrophobic ;
7 R-group of valine form different bonds / interactions with other amino acids ;
8 result in different tertiary structure / 3D conformation ;

[Effects of change of DNA sequence on haemoglobin, red blood cells and organism]

9 at low oxygen concentration, solubility of deoxygenated HbS decreases ;
10 HbS molecules stick to each other via their hydrophobic regions ;
11 polymerisation of the molecules into long fibres inside red blood cells ;
12 fibers of abnormal haemoglobin deform red blood cells into sickle shape ;

13 sickle-shaped RBCs clump to one another ;
14 sickle-shaped RBCs clog / block small capillaries ;
/ obtruct other cells from moving through the capillaries ;

15 sickle-shaped RBCs are ineffective in transporting oxygen gas ;
/ less oxygen carried / delivered to tissues ;
16 patients experience lethargy / tiredness ;
17 Painful crisis / "sickling"
/ pain in organ
/ organ damage ;

18 sickle-shaped RBCs have shorter life span compared to normal cells and hemolyse readily ;
(10-20 days versus 90-120 days)
19 anaemia observed in patients ;
(loss of RBCs cannot be compensated in time by rate of production of the bone marrow)

(b) Describe the development of cancer as a multistep process.

…………………………………………..……………………..…………………..…………[8]

1 loss of function mutation of (usually more than one) tumour suppressor gene ;
2 mutant tumour suppressor alleles act as recessive alleles / mutations must knock out both alleles in a cell's genome to produce abnormal protein ;
3 loss of arrest of cell cycle, loss of ability for DNA repair AND loss of ability to stimulate apoptosis ; (all 3 must be mentioned)

4 gain of function mutation of (at least) a proto-oncogene converts it into an oncogene ;
5 oncogenes act as dominant alleles / only need one mutated allele to produce abnormal protein ;
6 lead to uncontrolled cell division and excessive proliferation ;

7 subsequent accumulation of many mutations ;

8 <u>activation of telomerase gene</u> to produce telomerase enzyme ;
9 telomerase enzyme prevents the shortening of the chromosome ends ; (when telomere length reaches a critical length, important genes are lost. A normal cell will undergo apoptosis when critical length is reached)
10 cell can continue to divide indefinitely ;

11 cancer cells stimulate angiogenesis / formation of new network of blood vessels to the cancer cells ;
12 blood vessels provide the cancer cells oxygen and nutrients for growth AND to remove any waste products ;

13 loss of contact inhibition / density-dependence (and cells do not stop dividing) ;
14 loss of ability to differentiate ;
15 formation of benign tumour ;

16 cells in the tumour no longer exhibit anchorage dependence / loss of cell adhesion ;
17 metastasis / break loose and may enter the bloodstream / invade other tissues form secondary tumors ;
18 tumour is considered malignant.

(c) The activity of amylase can be measured by testing for the concentration of the glucose produced after the reaction.

Describe how the concentration of glucose produced can be measured using a colorimeter.

………………………………………...……………………..……………….………[5]

1 Prepare glucose standards / known concentrations of reducing sugar ;

2 Add equal volume of Benedict's solution to a test tube of the reducing sugar of unknown concentration ;
3 Place test tube in boiling water bath ;
4 Use the same volumes of solutions for both test and standard solutions ;

5 Change in colour of Benedict's solution from blue to green / yellow / orange / brown / brick red ;
6 Remove precipitate to obtain filtrate ;
CREDIT **description of method e.g. filtering / centrifuging and decanting**

7 Calibrate / zero, colorimeter ;
8 Using a blank / water / unreacted Benedict's solution ;
9 Read absorbance value of standards and unknown sugar solution ;

10 Obtain calibration curve from glucose standards / plot absorbance against reducing sugar concentration ;
11 Use reading of unknown sugar solution and read off graph to find concentration ;

- END OF PAPER -